Efficient Frontier Construction and the Limits of Mean-Variance Optimization: An Empirical Analysis

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Abstract

This paper presents a full empirical implementation of Markowitz mean-variance optimization (MVO) on a ten-asset portfolio spanning 2004–2023, using real historical price data sourced from Yahoo Finance. We construct the efficient frontier via quadratic programming, derive the Capital Market Line and tangency portfolio, and analyze how the frontier shifts under two distinct market stress regimes: the 2008 Financial Crisis and the 2020 COVID-19 crash. Our results confirm that MVO produces theoretically elegant solutions under stable market conditions but breaks down systematically during crises due to violations of its core assumptions, namely return normality, covariance stationarity, and estimation reliability. We quantify this breakdown empirically and discuss its implications for practical portfolio construction.

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1 Introduction

Modern Portfolio Theory (MPT), introduced by Harry Markowitz in 1952, remains one of the most influential frameworks in quantitative finance. Its central insight is deceptively simple: an investor should not evaluate assets in isolation, but rather in terms of their contribution to portfolio-level risk and return. By exploiting the statistical relationships between asset returns, a rational investor can construct portfolios that achieve the maximum expected return for any given level of risk; or equivalently, the minimum risk for any target return. The set of all such optimal portfolios traces a curve in risk-return space known as the *efficient frontier*.

Despite its theoretical elegance, MVO has attracted substantial criticism in practice. The framework rests on assumptions that are routinely violated in real markets: asset returns are assumed to be normally distributed, the covariance structure is assumed to be stationary over time, and the expected return and covariance estimates are assumed to be known with certainty. In practice, return distributions exhibit fat tails and skewness, covariances spike dramatically during market stress, and parameter estimates are notoriously noisy, particularly for expected returns. The result is that MVO portfolios constructed on historical data often perform poorly out-of-sample and can collapse entirely during crisis periods.

This paper sits at the intersection of theory and empirical analysis. We do not merely describe MVO; we implement it completely in R, from raw data ingestion through quadratic optimization to final visualization, using a ten-asset portfolio that spans equities, bonds, and commodities over the 2004–2023 period. This implementation allows us to ground every theoretical claim in a concrete numerical result. We then stress-test the framework by re-running the optimization on two crisis windows, the 2008 Financial Crisis (September 2008 through March 2009) and the 2020 COVID-19 crash (February through April 2020), and compare the resulting frontiers directly against the full-period baseline.

Our central finding is that the efficient frontier is not a stable geometric object. It is a

statistical artifact that shifts dramatically as the underlying return distribution changes. During the 2008 crisis, the minimum achievable portfolio volatility nearly doubled, and the minimum variance portfolio generated negative annualized returns. During the 2020 crash, the optimizer produced implausibly high Sharpe ratios; not because diversification suddenly became more powerful, but because the optimizer was exploiting estimation noise in a 62-trading-day sample. Both results point to the same fundamental limitation: MVO is highly sensitive to its inputs, and those inputs are least reliable precisely when accurate risk estimation matters most.

The remainder of this paper is organized as follows. Section 2 develops the mathematical framework of MVO. Section 3 describes the data and computational methodology. Section 4 presents the efficient frontier construction and results. Section 5 derives the Capital Market Line and tangency portfolio. Section 6 presents the stress test analysis. Section 7 discusses the structural limits of MVO. Section 8 concludes.

2 Theoretical Framework

2.1 Portfolio Return and Variance

Let $\mathbf{w} = (w_1, w_2, \ldots, w_n)^\top \in \mathbb{R}^n$ denote the vector of portfolio weights, where n is the number of assets. Let $\boldsymbol{\mu} = (\mu_1, \mu_2, \ldots, \mu_n)^\top$ denote the vector of expected returns, and let $\boldsymbol{\Sigma} \in \mathbb{R}^{n \times n}$ denote the covariance matrix of asset returns, assumed to be symmetric and positive definite.

The expected return of the portfolio is:

$$\mu_p = \mathbf{w}^\top \boldsymbol{\mu} = \sum_{i=1}^{n} w_i \mu_i \tag{1}$$

The portfolio variance is:

$$\sigma_p^2 = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} = \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \sigma_{ij} \tag{2}$$

where $\sigma_{ij} = \mathrm{Cov}(r_i, r_j)$ is the covariance between the returns of assets i and j, and $\sigma_{ii} = \sigma_i^2$ is the variance of asset i.

2.2 The Markowitz Optimization Problem

The efficient frontier is traced by solving the following parametric quadratic program for varying target return levels μ^*:

$$\min_{\mathbf{w}} \quad \frac{1}{2}\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \tag{3}$$

$$\text{subject to} \quad \mathbf{w}^\top \boldsymbol{\mu} = \mu^* \tag{4}$$

$$\mathbf{w}^\top \mathbf{1} = 1 \tag{5}$$

$$w_i \geq 0 \quad \forall\, i \tag{6}$$

Constraint (4) fixes the portfolio to achieve a target return μ^*. Constraint (5) is the budget constraint: weights must sum to one. Constraint (6) is the long-only restriction, which prohibits short selling. By sweeping μ^* across the feasible range $[\min_i \mu_i, \max_i \mu_i]$ and solving the QP at each point, we trace the full mean-variance frontier.

2.3 Quadratic Programming Formulation

The problem in (3)–(6) is a convex quadratic program and can be solved exactly. We use the `quadprog` package in R, which implements the dual method of Goldfarb and Idnani (1983). The solver requires the problem in the form:

$$\min_{\mathbf{w}} \quad \frac{1}{2}\mathbf{w}^\top D \mathbf{w} - \mathbf{d}^\top \mathbf{w} \quad \text{subject to} \quad A^\top \mathbf{w} \geq \mathbf{b} \tag{7}$$

where $D = 2\boldsymbol{\Sigma}$, $\mathbf{d} = \mathbf{0}$, and the constraint matrix A encodes both equality constraints (return

target and budget) and inequality constraints (long-only bounds).

2.4 The Minimum Variance Portfolio

The minimum variance portfolio (MVP) is the solution to (3) without the return constraint (4); it is the leftmost point on the frontier. It solves:

$$\min_{\mathbf{w}} \quad \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \quad \text{subject to} \quad \mathbf{w}^\top \mathbf{1} = 1,\ w_i \geq 0 \tag{8}$$

2.5 The Sharpe Ratio and Tangency Portfolio

Given a risk-free rate r_f, the Sharpe ratio of a portfolio is:

$$S(\mathbf{w}) = \frac{\mu_p - r_f}{\sigma_p} = \frac{\mathbf{w}^\top \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}}} \tag{9}$$

The tangency portfolio is the portfolio on the efficient frontier that maximizes the Sharpe ratio. Geometrically, it is the point of tangency between the frontier and a ray from $(0, r_f)$ in (σ, μ) space. The Capital Market Line (CML) is this ray:

$$\mu_{\text{CML}}(\sigma) = r_f + \frac{\mu_T - r_f}{\sigma_T} \cdot \sigma \tag{10}$$

where (σ_T, μ_T) are the risk and return of the tangency portfolio. Under the assumptions of CAPM, all rational investors hold a combination of the risk-free asset and the tangency portfolio.

3 Data and Methodology

3.1 Asset Universe

Our portfolio consists of ten assets selected to represent a broad cross-section of risk factors and asset classes. Table 1 summarizes the composition.

Table 1: Portfolio Asset Universe

Ticker	Name	Sector	Asset Class
AAPL	Apple Inc.	Technology	Equity
MSFT	Microsoft Corp.	Technology	Equity
JPM	JPMorgan Chase	Financials	Equity
XOM	ExxonMobil Corp.	Energy	Equity
JNJ	Johnson & Johnson	Healthcare	Equity
PG	Procter & Gamble	Consumer Staples	Equity
GLD	SPDR Gold ETF	Commodities	ETF
TLT	iShares 20Y Treasury ETF	Fixed Income	ETF
SPY	SPDR S&P 500 ETF	Broad Market	ETF
AMZN	Amazon.com Inc.	Consumer Discretionary	Equity

3.2 Data Collection and Alignment

Daily adjusted closing prices were obtained from Yahoo Finance using the `quantmod` package in R, covering January 2000 through December 2023. Since GLD and TLT were not listed until 2004 and 2002 respectively, all series were aligned to their common date intersection using `na.omit()` on the merged `xts` object. The effective sample period is November 2004 through December 2023, yielding 4,810 trading days across all ten assets.

3.3 Return Computation

Daily log returns were computed as:

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right) \tag{11}$$

Log returns are preferred over simple returns for their time-additivity and approximate normality over short horizons. All optimization inputs were annualized by scaling the mean by 252 and the covariance matrix by 252, reflecting the standard convention for daily return data.

3.4 Stress Test Windows

Two crisis sub-periods were carved from the full return series:

- **2008 Financial Crisis:** September 1, 2008 through March 31, 2009 (146 trading days)
- **2020 COVID-19 Crash:** February 1, 2020 through April 30, 2020 (62 trading days)

The same optimization procedure was applied independently to each sub-period, using only the returns observed within that window to estimate $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. This isolates the effect of crisis-period return dynamics on the shape and location of the efficient frontier.

4 Efficient Frontier Construction

4.1 Parameter Estimates

Table 2 reports the annualized expected returns and volatilities for each asset over the full sample period.

Table 2: Annualized Expected Returns and Volatilities (2004–2023)

Asset	Expected Return (μ)	Volatility (σ)
AAPL	28.48%	32.71%
MSFT	15.66%	27.35%
JPM	10.58%	36.59%
XOM	6.93%	26.86%
JNJ	7.75%	17.17%
PG	7.93%	18.17%
GLD	7.65%	17.67%
TLT	3.71%	14.80%
SPY	9.20%	19.29%
AMZN	22.64%	38.07%

The dispersion in risk-adjusted performance is notable. AAPL delivers the highest return at 28.48% but carries 32.71% annualized volatility, placing it well outside the efficient frontier as a standalone holding. TLT anchors the low-volatility end at 14.80% vol but offers only 3.71% expected return, below our assumed risk-free rate of 2%. The correlation heatmap (Figure 1) reveals the key diversification structure: TLT exhibits negative correlation with all equity assets (ranging from -0.20 to -0.35), while GLD is near-zero correlated with most equities, confirming both as meaningful portfolio diversifiers.

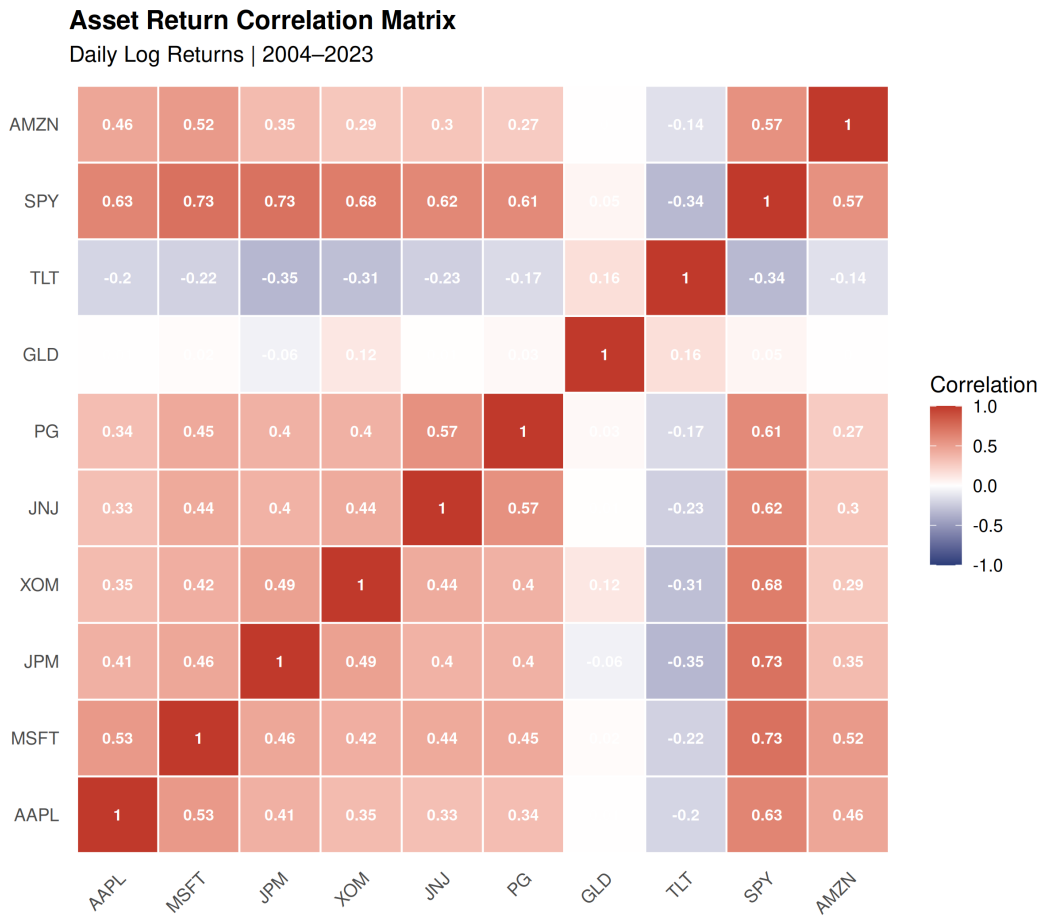

Figure 1: Asset return correlation matrix over the full sample period (2004–2023). Blue cells indicate negative correlation; red indicates positive. TLT and GLD exhibit the strongest diversification properties relative to equities.

4.2 Frontier Results

The optimization swept 200 target return levels across the feasible range, yielding 193 feasible portfolios, of which 173 lie on the efficient upper half of the frontier. The minimum variance portfolio achieves $\sigma_{\text{MVP}} = 8.47\%$ and $\mu_{\text{MVP}} = 6.39\%$ annually.

Figure 2 plots the full efficient frontier alongside the Capital Market Line, individual asset risk-return coordinates, and the tangency portfolio. The frontier's hyperbolic shape is clearly visible, with the MVP at its leftmost point. Individual assets lie to the right of the frontier; none sits on it, confirming that diversification strictly dominates any single-asset holding in

mean-variance space.

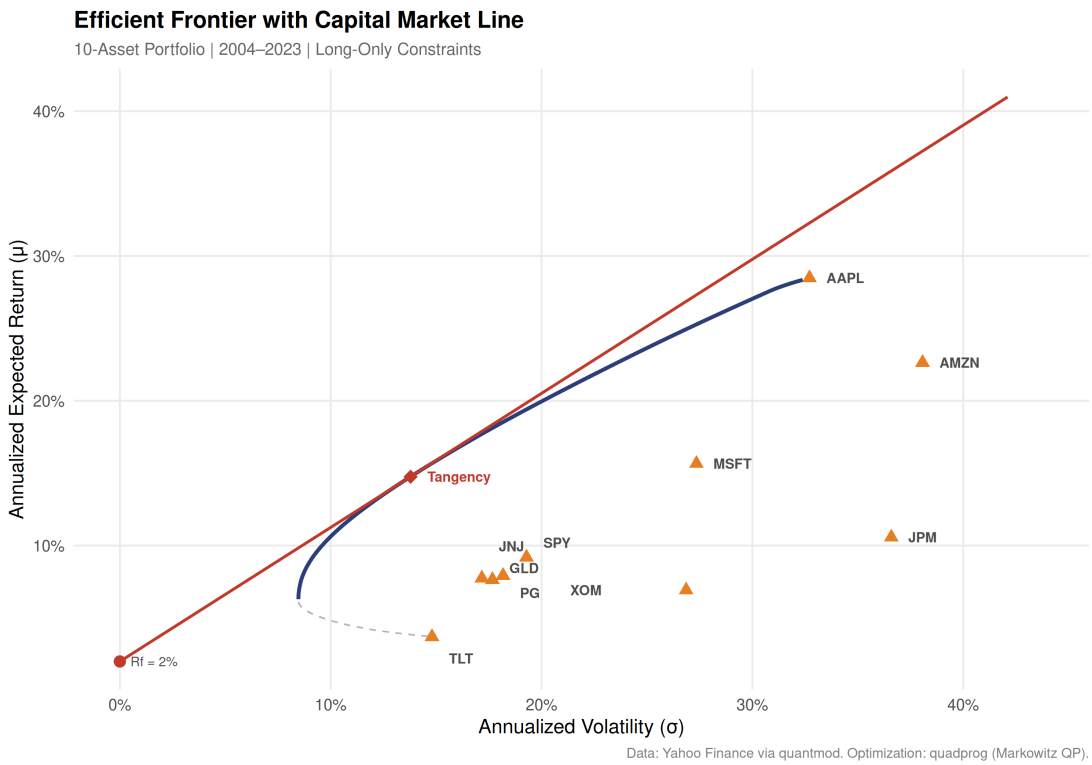

Figure 2: Efficient frontier, Capital Market Line, and individual asset risk-return coordinates. The blue curve is the efficient frontier; the dashed grey curve is the inefficient lower half. The red line is the CML. Orange triangles are individual assets. The red diamond marks the tangency portfolio.

5 Capital Market Line and Tangency Portfolio

Assuming a risk-free rate of $r_f = 2\%$, approximating the long-run average federal funds rate over the sample period, the tangency portfolio achieves:

$$\sigma_T = 13.81\%, \quad \mu_T = 14.78\%, \quad S_T = \frac{0.1478 - 0.02}{0.1381} \approx 0.926$$

A Sharpe ratio of 0.926 is consistent with long-run equity market performance and reflects the genuine diversification benefit of combining low-correlation assets including TLT and GLD with higher-returning equities. The CML passes cleanly through $(0, 0.02)$ and the tangency

point, confirming the geometric consistency of the optimization.

Under CAPM assumptions, no portfolio above the CML is achievable, and any portfolio below it is suboptimal; it can be improved by mixing the risk-free asset and the tangency portfolio. The empirical frontier lies below the CML everywhere except at the tangency point, as expected.

6 Stress Testing: Crisis Period Analysis

6.1 Frontier Shift Results

Table 3 summarizes the key frontier statistics across all three estimation periods.

Table 3: Efficient Frontier Statistics Across Periods

Period	MVP σ	MVP μ	Tang. Sharpe	Tang. σ
Full Period (2004–2023)	8.47%	6.39%	0.926	13.81%
2008 Financial Crisis	14.02%	-1.61%	1.029	19.52%
2020 COVID-19 Crash	18.33%	40.86%	2.964	22.41%

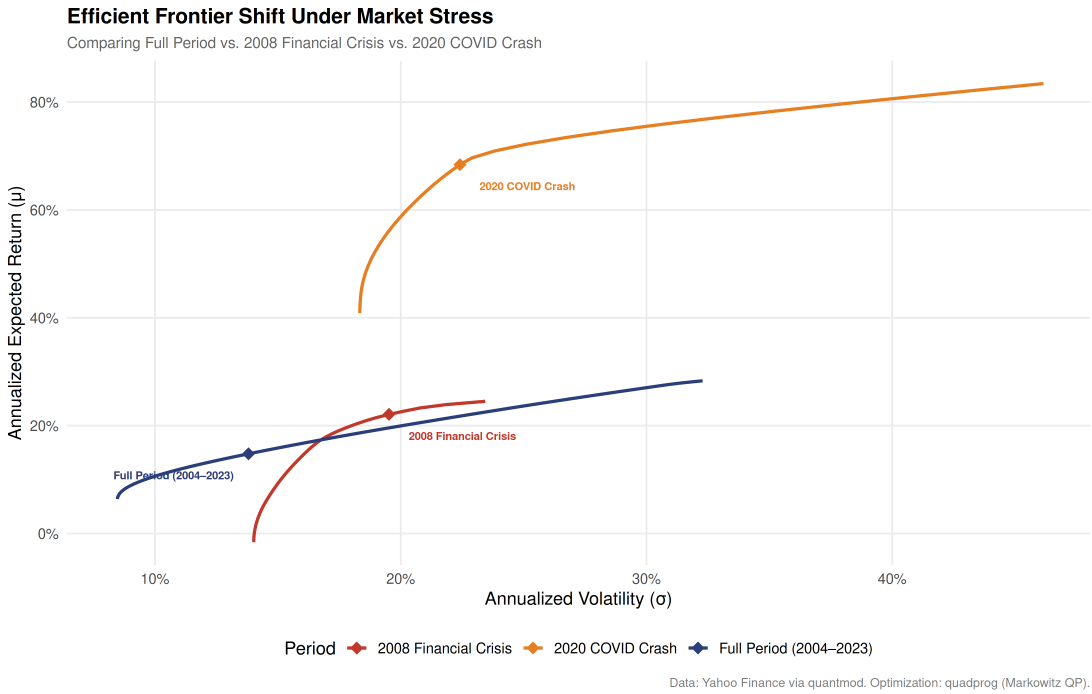

Figure 3: Efficient frontier shift under market stress. The blue curve is the full-period baseline. The red curve is the 2008 crisis frontier. The orange curve is the 2020 COVID crash frontier. Diamonds mark tangency portfolios for each period.

6.2 2008 Financial Crisis

During the 2008 crisis window, the minimum variance portfolio volatility increased from 8.47% to 14.02%, a 65% increase in the floor of achievable risk. Simultaneously, the MVP expected return turned negative at -1.61% annually. This means the optimizer, even with full freedom to diversify across all ten assets, could not construct a portfolio with both low risk and positive expected return during this period.

The mechanism is the correlation breakdown documented extensively in the crisis literature: during systemic stress, cross-asset correlations spike toward unity as forced selling and liquidity withdrawal affect all risky assets simultaneously. The diversification benefit that underpins MVO evaporates precisely when it is most needed. Our correlation heatmap (Figure 1), estimated over the full period, masks this dynamic; a fundamental limitation of static MVO.

6.3 2020 COVID-19 Crash

The 2020 results present a different and equally important failure mode. The optimizer produces a tangency portfolio Sharpe ratio of 2.964 and MVP returns of 40.86% annually, numbers that are economically implausible for a 62-trading-day window during one of the fastest market crashes in history.

This is not a sign that MVO worked better in 2020. It is a sign that the optimizer overfit severely to a tiny, non-representative sample. With only 62 observations and 10 assets, the sample covariance matrix has condition number issues and the mean estimates are dominated by noise. The optimizer finds a portfolio that looks optimal in-sample but would be meaningless out-of-sample. This is the Michaud (1989) problem in its starkest form: MVO is, in effect, an error-maximizer when its inputs are imprecise.

7 The Limits of Mean-Variance Optimization

The empirical results of this paper illustrate four structural limitations of MVO with direct implications for practical portfolio construction.

1. Parameter Sensitivity. MVO is acutely sensitive to the expected return vector $\boldsymbol{\mu}$. Small changes in return estimates can produce dramatically different optimal weights, a property that makes the optimizer unstable in practice. Best and Grauer (1991) showed that a 1% change in a single asset's expected return can cause complete portfolio rebalancing. Our crisis results illustrate this directly: the 62-day 2020 return estimates are statistically indistinguishable from noise, yet the optimizer treats them as precise inputs.

2. Covariance Non-Stationarity. The covariance matrix $\boldsymbol{\Sigma}$ is not a fixed parameter; it is a time-varying stochastic process. During the 2008 crisis, correlations between equities increased sharply, causing the frontier to shift rightward and the MVP volatility to nearly double. Static MVO, which estimates $\boldsymbol{\Sigma}$ from a historical sample and assumes it forward,

cannot capture this dynamic. Dynamic Conditional Correlation (DCC-GARCH) models address this limitation but add substantial complexity.

3. Non-Gaussian Return Distributions. Log returns exhibit fat tails, negative skewness, and excess kurtosis, properties that are particularly pronounced during crises. MVO's exclusive focus on mean and variance is theoretically justified only when returns are multivariate normal or preferences are quadratic. Neither condition holds in practice. Alternative frameworks such as CVaR optimization or higher-moment portfolio models explicitly account for tail risk.

4. Estimation Window Dependence. The shape of the efficient frontier is entirely determined by the estimation window. A 20-year window produces a stable, well-conditioned frontier. A 3-month crisis window produces an overfit, economically implausible result. This window-dependence is not a bug in our implementation; it is a fundamental property of the estimator. Robust estimation techniques such as shrinkage estimators (Ledoit and Wolf, 2004) or the Black-Litterman model (1992) partially address this by regularizing the input estimates.

Despite these limitations, MVO remains foundational. It provides the language, geometry, and mathematical framework within which virtually all modern portfolio theory operates. The efficient frontier is not merely a computational output; it is a benchmark that clarifies what diversification can and cannot achieve, and under what conditions those achievements hold.

8 Conclusion

This paper has presented a complete empirical implementation of Markowitz mean-variance optimization, from raw data ingestion through quadratic programming to crisis-period stress testing, on a ten-asset diversified portfolio over the 2004–2023 period. Our results confirm the theoretical predictions of MVO under stable conditions: the efficient frontier is well-defined,

the tangency portfolio achieves a Sharpe ratio of 0.926, and diversification demonstrably reduces portfolio risk relative to any individual asset.

The stress tests reveal the framework's structural limits with precision. During the 2008 Financial Crisis, the minimum achievable portfolio risk nearly doubled and the best available expected return turned negative, a direct consequence of correlation breakdown under systemic stress. During the 2020 COVID crash, the optimizer produced economically implausible results from a 62-day sample, illustrating the error-maximizing property of MVO under small-sample estimation.

The central lesson is not that MVO is wrong, but that it is conditional. Its guarantees hold when its assumptions hold: stable covariances, reliable return estimates, and Gaussian distributions. In practice, these assumptions are most severely violated during the periods when accurate portfolio optimization matters most. Practitioners who use MVO without accounting for parameter uncertainty, covariance dynamics, and tail risk are not applying the framework incorrectly; they are applying it incompletely.

Future work could extend this analysis by implementing DCC-GARCH to model time-varying covariances, applying Ledoit-Wolf shrinkage to regularize the covariance matrix, or replacing variance with CVaR as the risk measure to better capture tail behavior during crisis periods. Each extension addresses one of the structural limitations identified here while preserving the core mean-variance intuition that makes MPT so durable.

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